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Anatomy and Physiology — Semester 1 Teacher Manual

Florida course 2000350 · For the teacher; not for students

> **Read this first.** This manual describes what is in the Semester 1 kit, how it is meant to be run, and what you need to set up in Canvas after importing it. Everything here is drawn from the course pages themselves. Where the course leaves a decision to you (grade weights, late-work policy, due dates), this manual says so.

1. What this course is

Anatomy and Physiology (Florida 2000350) for grades 11–12, taught fully asynchronously in Canvas. Every lesson pairs structure with function and states cross-system connections explicitly. Weekly VR work uses ENGAGE with the HoloAnatomy environment: students sculpt and label the structures they are studying and reassemble them into a module-end figure. Semester 1 has seven modules and a finals week (17 weeks); Semester 2 continues with Modules 8–12.

2. Modules at a glance

Module	System	When	Quarter	Virtue thread	Lesson time
1	Introduction to Anatomy & Physiology	Week 1	Q1	Wonder	315 min per week (4 lessons)
2	Tissues & the Integumentary System	Week 2	Q1	Wonder	317 min per week (4 lessons)
3	Skeletal System	Weeks 3–5	Q1	Perseverance	330 min, 310 min, 340 min per week (4 lessons)
4	Muscular System	Weeks 6–8	Q1	Self-government	358 min, 364 min, 365 min per week (4 lessons)

5	Nervous System	Weeks 9–11	Q2	Courage	350 min, 355 min, 355 min per week (4 lessons)
6	Endocrine System	Weeks 12–13	Q2	Courtesy	340 min, 350 min per week (4 lessons)
7	Cardiovascular System	Weeks 14–16	Q2	Service	350 min, 355 min, 360 min per week (4 lessons)

Lesson time is the sum of the four lessons' stated activity minutes for each week.

3. How a week is built

Each full week has four readings, four paired lessons, four paired assignments and one reading quiz. The reading carries the content; the lesson page carries the activities (2–4 per lesson), the VR session description and the recording or sketching tasks; the assignment page carries the focus question, directions, success criteria, rubric and submission instructions. Each module also has a module writing assignment and a module-end VR snapshot. Module 1 and Module 2 are one-week modules. Module 2 has a fifth, optional assignment (Cancer: When Cell Division Breaks Down, 8 points), set to *omit from final grade* so you can decide whether it counts.

Pages in the kit are wrappers that display the lesson, reading and assignment pages hosted by Optima. Students work on the assignment pages and submit through the Canvas assignment of the same name.

4. Grading

Category (Canvas assignment group)	Items	Points	Graded by
Lesson Assignments	64	784	Teacher, rubric on each page
Module Writing Assignments	7	188	Teacher, rubric on each page
VR Snapshot Submissions	7	155	Teacher, rubric on each page
Reading Quizzes	16	138	Automatic
Semester Finals	2	74	Review Activity: teacher; Final: automatic
Total	**96**	**1339**	

Weighting is yours to set. The kit ships unweighted (all groups at 0%). Set the group weights in the Course Kit widget or the Canvas gradebook to match your school's policy.

Typical point values. Lesson assignments are usually 12 points (range 10–16); module writing assignments are 18–30 points; VR snapshots are 12–25 points. Each rubric is printed at the bottom of its assignment page, and rubric rows sum to the stated total.

Quizzes. The kit contains 16 reading quizzes (138 questions in total, 1 point each) and the finals quiz. All are multiple choice, true/false or matching, so all are auto-graded. Each reading quiz allows one attempt with shuffled answers and shows the correct answers afterward. Answer keys are part of the kit.

Review Activity (Finals Week). The page itself says it is auto-checked for completion and that the teacher spot-reviews one or two cells per student plus the reflection paragraph.

5. VR and recorded work

- **VR.** Weekly sessions use ENGAGE with HoloAnatomy; Module 3 onward the course defaults to the male figure through Module 11. Students save their work between sessions and carry the figure forward from module to module. Each module ends with a VR snapshot (an IFX file plus a short caption; an audio caption is acceptable where the VR setup supports it). Module 1's VR sessions are orientation only, so its snapshot is a lighter 12-point submission. Students are told to contact you before the due date if they cannot capture or export a file.
- **Recorded narration.** Many activities ask students to talk through an idea aloud and record it, rather than write it. This is deliberate: it makes the work harder to outsource. Where an assignment requires a recording, the assignment page says what to submit and in what format.

6. Finals week

Finals Week has the Review Lesson, the Review Activity (24 points) and the Semester 1 Comprehensive Final (50 questions, 50 points). The final draws proportionally from every module's reading-quiz bank: 2 questions from Module 1, 3 from Module 2, 10 each from Modules 3, 4 and 5, 6 from Module 6 and 9 from Module 7. The bank is 138 questions in all.

> **Set these on the final after import.** (1) The kit cannot carry a time limit, so set one yourself; the course design calls for 90 minutes. (2) The kit cannot carry a question-group draw, so the final ships as one fixed set of 50 questions chosen evenly through each module's bank. If you want each student to see a different draw, build seven question banks from the reading quizzes and draw 2/3/10/10/10/6/9. (3) The course design asks that the Review Activity be submitted before the final opens; set that as a Canvas module prerequisite. (4) The final ships with correct answers hidden; the course design shows them only after the quiz closes for the class, so switch that on once everyone has finished.

7. Before you launch: a checklist

1. Import the kit as a **Canvas Course Export Package**, and leave "Import existing quizzes as New Quizzes" **unchecked**; ticking it empties ungraded quizzes.
2. Use the Course Kit widget to write your home page, set the term start so due dates are placed by week, and choose gradebook weights.
3. Set the final's time limit and the Review Activity prerequisite (section 6).
4. Confirm your students have ENGAGE access and a way to record audio before Week 1.
5. Publish the modules when you are ready; the Teacher Resources module is unpublished by design.

8. Module-by-module reference

Items appear in student order. Points are shown for assignments and quizzes; each reading-quiz question is 1 point.

Module 1 — Introduction to Anatomy & Physiology

Item	Type	Points
M1 Writing Assignment — From Atoms to You	Assignment	18
Week 1		
Reading 1.01 — What is Anatomy & Physiology? Levels of Organization	Page	
Lesson 1.01 — What is Anatomy & Physiology? Levels of Organization	Page	
Lesson 1.01 — Levels of Organization Reflection	Assignment	12
Reading 1.02 — Anatomical Position, Planes, Regions, Cavities	Page	
Lesson 1.02 — Anatomical Position, Planes, Regions, Cavities	Page	
Lesson 1.02 — Directional Vocabulary in Practice	Assignment	12
Reading 1.03 — The Chemistry of Life	Page	
Lesson 1.03 — The Chemistry of Life	Page	
Lesson 1.03 — Macromolecules on the Label	Assignment	12
Reading 1.04 — Enzymes — Biological Catalysts	Page	
Lesson 1.04 — Enzymes — Biological Catalysts	Page	
Lesson 1.04 — Enzymes Under Physiological Conditions	Assignment	12
Module 1, Week 1 — Reading Quiz	Quiz	6
M1 VR Snapshot — Your First Orientation in the Body	Assignment	12

Module 2 — Tissues & the Integumentary System

Item	Type	Points
M2 Writing Assignment — Four Tissues, One Skin	Assignment	20
Week 2		
Reading 2.01 — Epithelial and Connective Tissues	Page	
Lesson 2.01 — Epithelial and Connective Tissues	Page	
Lesson 2.01 — Tissue Identification & Form-Function	Assignment	12
Reading 2.02 — Muscle and Nervous Tissues	Page	
Lesson 2.02 — Muscle and Nervous Tissues	Page	
Lesson 2.02 — Excitability: How Muscles and Neurons Work	Assignment	12
Reading 2.03 — Integument, Day 1	Page	
Lesson 2.03 — Integument, Day 1 of 2	Page	
Lesson 2.03 — Skin's Layers: A Form-Function Start	Assignment	10
Reading 2.04 — Integument, Day 2	Page	
Lesson 2.04 — Integument, Day 2 of 2	Page	
Lesson 2.04 — Skin as a Form-Function Case Study	Assignment	12
Lesson 2.04 — Cancer: When Cell Division Breaks Down (Optional)	Assignment	8
Module 2, Week 1 — Reading Quiz	Quiz	8
M2 VR Snapshot — The Assembled Integument	Assignment	18

Module 3 — Skeletal System

Item	Type	Points
M3 Writing Assignment — Synovial Joints: Form, Function, and What Breaks	Assignment	30
Week 3		
Reading 3.01 — Bone Biology — Tissue, Shapes, and Markings	Page	
Lesson 3.01 — Bone Biology — Tissue, Shapes, and Markings	Page	
Lesson 3.01 — Bone Shapes & Markings Field Notes	Assignment	12
Reading 3.02 — The Skull — Cranial Bones	Page	
Lesson 3.02 — The Skull — Cranial Bones	Page	
Lesson 3.02 — Label the Cranial Vault	Assignment	12

Reading 3.03 — Facial Bones, the Hyoid, and Dentition	Page	
Lesson 3.03 — Facial Bones, the Hyoid, and Dentition	Page	
Lesson 3.03 — Face, Hyoid & Dentition Quick-Check	Assignment	12
Reading 3.04 — The Vertebral Column — Overview, Cervical, Thoracic	Page	
Lesson 3.04 — The Vertebral Column — Overview, Cervical, Thoracic	Page	
Lesson 3.04 — Atlas, Axis, and the Twelve Ribs	Assignment	12
Module 3, Week 1 — Reading Quiz	Quiz	10
Week 4		
Reading 3.05 — The Lower Spine — Lumbar, Sacrum, Coccyx, and Intervertebral Discs	Page	
Lesson 3.05 — The Lower Spine	Page	
Lesson 3.05 — The Disc and the Load	Assignment	12
Reading 3.06 — The Thorax	Page	
Lesson 3.06 — The Thorax	Page	
Lesson 3.06 — Building the Cage that Breathes	Assignment	12
Reading 3.07 — The Shoulder Girdle and Upper Arm	Page	
Lesson 3.07 — Shoulder Girdle and Upper Arm	Page	
Lesson 3.07 — Mobility vs. Stability at the Shoulder	Assignment	12
Reading 3.08 — Forearm, Wrist, and Hand	Page	
Lesson 3.08 — Forearm, Wrist, and Hand	Page	
Lesson 3.08 — The Hand You Are Using to Read This	Assignment	12
Module 3, Week 2 — Reading Quiz	Quiz	10
Week 5		
Reading 3.09 — Pelvic Girdle and Femur	Page	
Lesson 3.09 — Pelvic Girdle and Femur	Page	
Lesson 3.09 — The Strongest Bone	Assignment	12
Reading 3.10 — Lower Leg, Ankle, and Foot	Page	
Lesson 3.10 — Lower Leg, Ankle, and Foot	Page	
Lesson 3.10 — The Footprint Architecture	Assignment	12
Reading 3.11 — Joints	Page	

Lesson 3.11 — Joints	Page	
Lesson 3.11 — The Architecture of Movement	Assignment	16
Reading 3.12 — Skeletal Pathology and Synthesis	Page	
Lesson 3.12 — Skeletal Pathology and Synthesis — Module-End Reassembly	Page	
Lesson 3.12 — The Skeleton in One Picture	Assignment	12
Module 3, Week 3 — Reading Quiz	Quiz	8
M3 VR Snapshot — The Assembled Skeleton	Assignment	25

Module 4 — Muscular System

Item	Type	Points
M4 Writing Assignment — From Nerve Impulse to Muscle Fiber Shortening	Assignment	30
Week 6		
Reading 4.01 — Three Muscle Types — Skeletal, Smooth, and Cardiac	Page	
Lesson 4.01 — Three Muscle Types — Skeletal, Smooth, and Cardiac	Page	
Lesson 4.01 — Three Muscle Types Field Notes	Assignment	12
Reading 4.02 — Skeletal Muscle Microscopic Anatomy	Page	
Lesson 4.02 — Skeletal Muscle Microscopic Anatomy	Page	
Lesson 4.02 — Muscle Fiber Infographic	Assignment	12
Reading 4.03 — The Sarcomere and the Sliding Filament Theory	Page	
Lesson 4.03 — The Sarcomere and the Sliding Filament Theory	Page	
Lesson 4.03 — The Cross-Bridge Cycle in Sequence	Assignment	14
Reading 4.04 — The Neuromuscular Junction	Page	
Lesson 4.04 — The Neuromuscular Junction	Page	
Lesson 4.04 — The Chain of Command at the NMJ	Assignment	14
Module 4, Week 1 — Reading Quiz	Quiz	9
Week 7		
Reading 4.05 — Muscles of the Head and Neck	Page	
Lesson 4.05 — Muscles of the Head and Neck	Page	
Lesson 4.05 — Head & Neck Muscle Atlas	Assignment	12
Reading 4.06 — Muscles of the Back and Thorax	Page	

Lesson 4.06 — Muscles of the Back and Thorax	Page	
Lesson 4.06 — Back & Thorax Muscle Atlas	Assignment	14
Reading 4.07 — Muscles of the Abdominal Wall	Page	
Lesson 4.07 — Muscles of the Abdominal Wall	Page	
Lesson 4.07 — Abdominal Wall Muscle Atlas	Assignment	12
Reading 4.08 — Muscles of the Upper Limb	Page	
Lesson 4.08 — Muscles of the Upper Limb	Page	
Lesson 4.08 — Upper Limb Muscle Atlas	Assignment	12
Module 4, Week 2 — Reading Quiz	Quiz	10
Week 8		
Reading 4.09 — Muscles of the Pelvis and Lower Limb	Page	
Lesson 4.09 — Muscles of the Pelvis and Lower Limb	Page	
Lesson 4.09 — Lower Limb Muscle Atlas	Assignment	12
Reading 4.10 — Muscle Metabolism	Page	
Lesson 4.10 — Muscle Metabolism	Page	
Lesson 4.10 — Three Energy Systems in Action	Assignment	14
Reading 4.11 — Exercise Physiology	Page	
Lesson 4.11 — Exercise Physiology	Page	
Lesson 4.11 — Training and the Body You Will Live In	Assignment	14
Reading 4.12 — Homeostatic Imbalances of the Muscular System	Page	
Lesson 4.12 — Homeostatic Imbalances of the Muscular System	Page	
Lesson 4.12 — Muscles in Motion	Assignment	16
Module 4, Week 3 — Reading Quiz	Quiz	7
M4 VR Snapshot — The Assembled Muscular System	Assignment	25

Module 5 — Nervous System

Item	Type	Points
M5 Writing Assignment — The Reflex That Saves You Before You Know It	Assignment	30
Week 9		

Reading 5.01 — Overview of the Nervous System — CNS, PNS, Neurons, and Glia	Page	
Lesson 5.01 — Overview of the Nervous System	Page	
Lesson 5.01 — CNS / PNS Map and Neuron Field Notes	Assignment	12
Reading 5.02 — The Neuron at Work	Page	
Lesson 5.02 — The Neuron at Work	Page	
Lesson 5.02 — The Action Potential Step-by-Step	Assignment	12
Reading 5.03 — The Synapse	Page	
Lesson 5.03 — The Synapse	Page	
Lesson 5.03 — Label the Synapse and Trace a Signal	Assignment	12
Reading 5.04 — Reflex Arcs	Page	
Lesson 5.04 — Reflex Arcs	Page	
Lesson 5.04 — Diagram a Reflex Arc	Assignment	12
Module 5, Week 1 — Reading Quiz	Quiz	10
Week 10		
Reading 5.05 — The Brain — Four Major Divisions	Page	
Lesson 5.05 — The Brain — Four Major Divisions	Page	
Lesson 5.05 — Four Brain Divisions Field Notes	Assignment	12
Reading 5.06 — Cerebrum — Cortex, Lobes, Ventricles, Meninges	Page	
Lesson 5.06 — Cerebrum — Cortex, Lobes, Ventricles, Meninges	Page	
Lesson 5.06 — Lobes, Ventricles, Meninges	Assignment	12
Reading 5.07 — The Spinal Cord	Page	
Lesson 5.07 — The Spinal Cord	Page	
Lesson 5.07 — Label the Spinal Cord Cross-Section	Assignment	12
Reading 5.08 — Cranial and Spinal Nerves	Page	
Lesson 5.08 — Cranial and Spinal Nerves	Page	
Lesson 5.08 — Map the 12 Cranial Nerves	Assignment	12
Module 5, Week 2 — Reading Quiz	Quiz	9
Week 11		
Reading 5.09 — The Autonomic Nervous System	Page	

Lesson 5.09 — The Autonomic Nervous System	Page	
Lesson 5.09 — ANS Comparison and Scenarios	Assignment	12
Reading 5.10 — The Eye and Vision	Page	
Lesson 5.10 — The Eye and Vision	Page	
Lesson 5.10 — Label the Eye	Assignment	12
Reading 5.11 — The Ear, Taste, and Smell	Page	
Lesson 5.11 — The Ear, Taste, and Smell	Page	
Lesson 5.11 — Label the Ear and the Other Special Senses	Assignment	12
Reading 5.12 — Module Synthesis	Page	
Lesson 5.12 — Module Synthesis	Page	
Lesson 5.12 — The Somatosensory Pathway Trace	Assignment	12
Module 5, Week 3 — Reading Quiz	Quiz	9
M5 VR Snapshot — The Assembled Nervous System	Assignment	25

Module 6 — Endocrine System

Item	Type	Points
M6 Writing Assignment — Two Speeds of Stress	Assignment	30
Week 12		
Reading 6.01 — Endocrine vs. Exocrine: What Is a Hormone?	Page	
Lesson 6.01 — Endocrine vs. Exocrine: What Is a Hormone?	Page	
Lesson 6.01 — Hormone Family ID	Assignment	12
Reading 6.02 — Endocrine vs. Neural Control of Physiology	Page	
Lesson 6.02 — Endocrine vs. Neural Control of Physiology	Page	
Lesson 6.02 — Neural vs. Endocrine Case Study	Assignment	12
Reading 6.03 — The Hypothalamus and the Pituitary Gland	Page	
Lesson 6.03 — The Hypothalamus and the Pituitary Gland	Page	
Lesson 6.03 — Trace an HPT, HPA, or HPG Axis	Assignment	12
Reading 6.04 — The Thyroid and Parathyroid Glands	Page	
Lesson 6.04 — The Thyroid and Parathyroid Glands	Page	
Lesson 6.04 — Calcium Homeostasis Walkthrough	Assignment	12

Module 6, Week 1 — Reading Quiz	Quiz	8
Week 13		
Reading 6.05 — The Adrenal Glands	Page	
Lesson 6.05 — The Adrenal Glands	Page	
Lesson 6.05 — Layered Stress Response	Assignment	12
Reading 6.06 — The Pancreas — Insulin, Glucagon, and Blood Glucose Homeostasis	Page	
Lesson 6.06 — The Pancreas	Page	
Lesson 6.06 — The Insulin-Glucagon See-Saw	Assignment	12
Reading 6.07 — Secondary Endocrine Organs	Page	
Lesson 6.07 — Secondary Endocrine Organs	Page	
Lesson 6.07 — Secondary Endocrine Roundup	Assignment	12
Reading 6.08 — Synthesis — The Chemistry of Coordination	Page	
Lesson 6.08 — Synthesis — The Chemistry of Coordination	Page	
Lesson 6.08 — Disorder-to-Mechanism Map	Assignment	12
Module 6, Week 2 — Reading Quiz	Quiz	8
M6 VR Snapshot — The Assembled Endocrine Map	Assignment	25

Module 7 — Cardiovascular System

Item	Type	Points
M7 Writing Assignment — Pressure, Resistance, and the Damage of Hypertension	Assignment	30
Week 14		
Reading 7.01 — What Is Blood? Plasma and the Formed Elements	Page	
Lesson 7.01 — What Is Blood? Plasma and the Formed Elements	Page	
Lesson 7.01 — The Centrifuge Tube	Assignment	12
Reading 7.02 — Red Blood Cells: Hemoglobin and Oxygen Transport	Page	
Lesson 7.02 — Red Blood Cells: Hemoglobin and Oxygen Transport	Page	
Lesson 7.02 — Form Follows Function: The Red Blood Cell	Assignment	12
Reading 7.03 — Leukocytes and Platelets	Page	

Lesson 7.03 — Leukocytes and Platelets	Page	
Lesson 7.03 — The Leukocyte ID Sheet	Assignment	12
Reading 7.04 — Hemostasis and Blood Typing	Page	
Lesson 7.04 — Hemostasis and Blood Typing	Page	
Lesson 7.04 — Transfusion Decisions	Assignment	12
Module 7, Week 1 — Reading Quiz	Quiz	9
Week 15		
Reading 7.05 — External Heart Anatomy	Page	
Lesson 7.05 — The Heart from the Outside	Page	
Lesson 7.05 — The Six Great Vessels	Assignment	12
Reading 7.06 — Inside the Heart	Page	
Lesson 7.06 — Inside the Heart	Page	
Lesson 7.06 — Inside the Pump	Assignment	12
Reading 7.07 — The Cardiac Cycle	Page	
Lesson 7.07 — Cardiac Cycle & Conduction	Page	
Lesson 7.07 — The Conduction Pathway Trace	Assignment	12
Reading 7.08 — Coronary Circulation	Page	
Lesson 7.08 — Coronary Circulation	Page	
Lesson 7.08 — The Coronary Map	Assignment	12
Module 7, Week 2 — Reading Quiz	Quiz	9
Week 16		
Reading 7.09 — Major Arteries	Page	
Lesson 7.09 — Major Arteries	Page	
Lesson 7.09 — The Arterial Tree	Assignment	12
Reading 7.10 — Major Veins	Page	
Lesson 7.10 — Major Veins	Page	
Lesson 7.10 — The Venous Return	Assignment	12
Reading 7.11 — Blood Flow, Pressure, and Resistance	Page	
Lesson 7.11 — Blood Flow, Pressure, and Hypertension	Page	

Lesson 7.11 — The Pressure Lab	Assignment	12
Reading 7.12 — Fetal Circulation	Page	
Lesson 7.12 — Fetal Circulation	Page	
Lesson 7.12 — The First Breath	Assignment	12
Module 7, Week 3 — Reading Quiz	Quiz	8
M7 VR Snapshot — The Assembled Cardiovascular System	Assignment	25

Semester 1 — Finals Week

Item	Type	Points
Week 17 — Finals Week		
Review Lesson — Semester 1 Review	Page	
S1 Review Activity — Six-System Comparative Chart	Assignment	24
Semester 1 Comprehensive Final — Anatomy & Physiology	Quiz	50

9. Notes and known limits

- Videos and animations suggested by the subject-matter reviewers are referenced in some readings but are not hosted in the course.
- One figure (Adrenal Cortex vs. Adrenal Medulla, Reading 6.05) has no real source and is intentionally absent; a comment in the page source marks where it goes once a properly sourced image is available.
- Readings credit images to their original sources beneath each figure.